

Youtube

Functional requirements:

user should be able to view any video based on his preference,
user should be able to search for any video based on his preference,
user should be able to subscribe any channel based on his preference,

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

user,
video,
channel

API :

POST:/upload

```
{  
  userID,  
  {filemetadata}  
}
```

GET:/search/{videoID}

HLD :

